WORKSHEET 1.12

Unit 1 Free-Response Practice

Unit 1: Limits and Continuity · two questions, 9 points each · suggested time 30 minutes

Instructions

- Show all your work. A correct answer with no supporting work may not earn credit.
- Use correct notation: "lim" stays until you substitute, and "=" joins only equal things.
- When asked to justify, name the theorem and state its conditions.
- Question 1: no calculator. Question 2: a graphing calculator is required; give decimals to three places.

Question 1 · No calculator

Let f be the function defined by

$$f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2}, & x < 2 \\ k, & x = 2 \\ \sqrt{x + 7}, & x > 2 \end{cases}$$

where k is a constant.

(a) Find $\lim\limits_{x \to 2^-} f(x)$ and $\lim\limits_{x \to 2^+} f(x)$. Show the work that leads to your answers.

(b) For what value of k is f continuous at $x = 2$? Justify your answer.

(c) Suppose instead that $k = 5$. What type of discontinuity does f have at $x = 2$? Give a reason.

(d) Evaluate $\lim\limits_{x \to \infty} \dfrac{f(x)}{\sqrt{x}}$.

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(e) Let k have the value found in part (b). Is there a number c, $2 < c < 9$, such that $f(c) = 3.5$? Justify your answer.

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Question 2 · Graphing calculator required

The amount of water in a reservoir, $W(t)$ megalitres, is a continuous function of time t days. Selected values are shown.

t	0	2	6	10
$W(t)$	40	52	46	61

(a) Find the average rate of change of W on the interval $2 \le t \le 6$. Using correct units, interpret your answer in context.

(b) What is the least number of times t in $0 \le t \le 10$ at which $W(t) = 50$? Justify your answer.

(c) An engineer models the reservoir with $M(t) = 60 - 20e^{-0.3t}$. Find $\lim_{t \to \infty} M(t)$ and interpret it in context.

(d) Use the model to estimate $\lim_{t \to 0} \dfrac{M(t) - 40}{t}$ to three decimal places, showing the values you used.

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